# Some Basic Probability

Probability and Random Variables

- Discrete (vs. Continuous) RV
- Gambler's Fallacy
- Independent vs. Dependent Events

Discrete Random Variables

- Probability Function (p.f.)
- Distribution Function (d.f.)
- Expected Value

# Learning to use R

## Outline

I. Recap

II. Data Manipulation
- Indexing
- Data Frame
- Data Input/Output

III. Plotting with R
- Line Charts
- Histogram
- Boxplot
- Violin plot
- Saving Plots

# Height of Martians – The Population

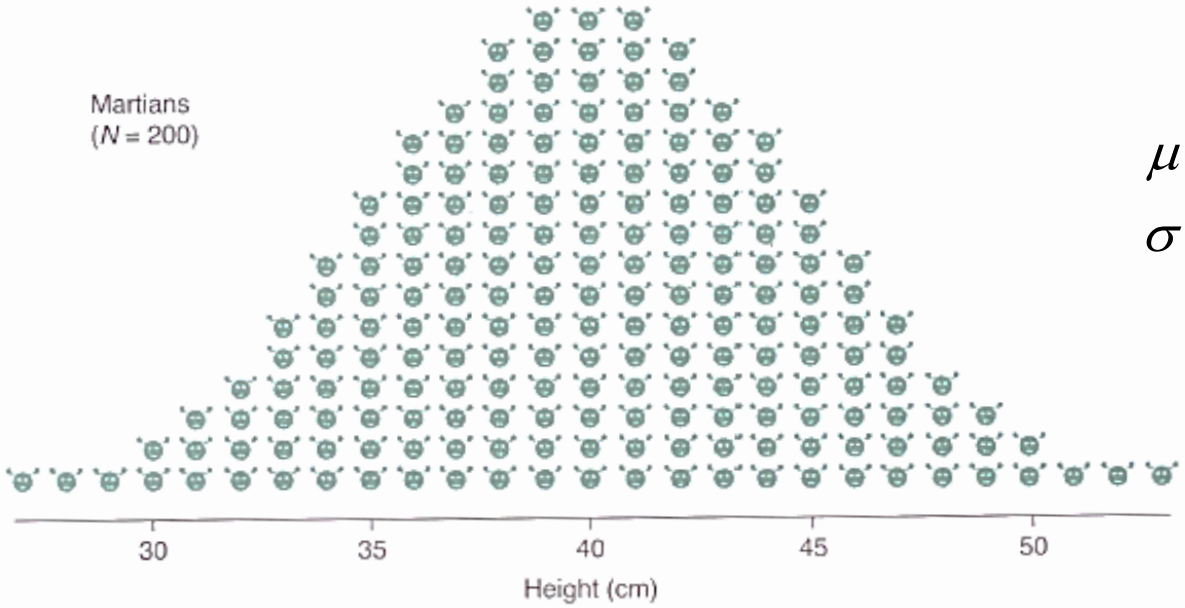



**FIGURE 2-1.** Distribution of heights of 200 Martians, with each Martian's height represented by a single point. Notice that any individual Martian is more likely to have a height near the mean height of the population (40 cm) than far from it and is equally likely to be shorter or taller than average.

$$\mu = 40.cm$$

$$\sigma = 5.0cm$$

$$x_i \ , \ i = 1, \ldots, \ N$$

Central Tendency - Mean

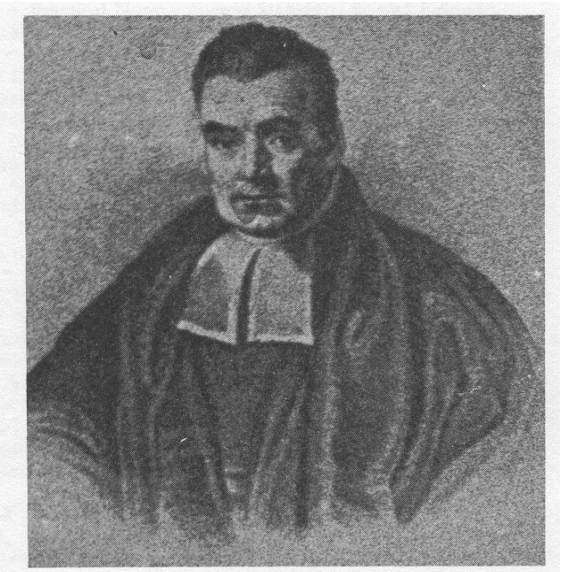
$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Dispersion - Variance

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - \mu\right)^2$$

# Early Contributors to the Theory of Probability

A gambler's dispute in 1654 led to the creation of a mathematical theory of probability by Blaise Pascal and Pierre de Fermat.

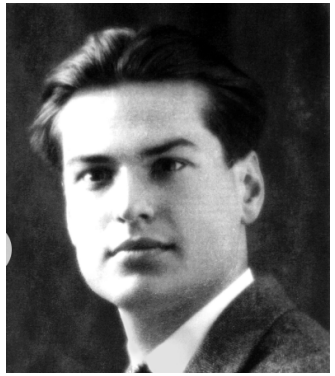


Reverend Thomas Bayes (1702-1761)
1763: *An Essay Towards Solving a Problem in the Doctrine of Chances*



Pierre de Laplace (1749-1827)
1812: *Théorie Analytique des Probabilités*

'Our world, our life, our destiny, are dominated by Uncertainty; this is perhaps the only statement we may assert without uncertainty'



Bruno de Finetti. Insurance and the views about probability.
Giornale dell'Istituto Italiano degli Attuari 1979; **50**:71–83.

# Some Basic Probability

**Random Process:** Any process for which all possible results are known but for which the *actual* results cannot be predicted with certainty in advance

**Random Variable (RV):** Outcome of a random process

**Experiment:** A procedure used to generate outcomes (or make measurements) from a random process

**Sample Space** ($D$): The set of all possible outcomes from an experiment

# Examples of Random Processes/Variables/Experiments



- Toss a single coin; record H or T

- Select a student from BME 423 class and measure HR

- Toss 3 coins, record the total number of heads



# Probability

The **probability** of an event is its relative frequency:

- $A$ = some event
- N($A$) = count of all outcomes in this event
- $D$ = sample space (all possible outcomes of experiment)
- N($D$) = count of all possible outcomes of experiment
- P($A$) = N($A$) / N($D$): probability of $A$'s occurring

# Discrete Random Variable

Definition: RV that can take on one of only a finite set of values

Example 1: Outcome of a coin toss – H or T

Represent the RV by $X$ and its value by $x$

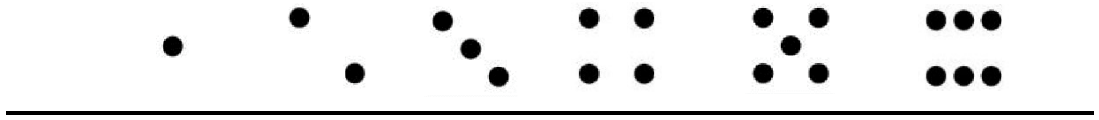
| | H | T |
|---|---|---|
| $x$ | 1 | 2 |

Prob $(X = x)$ ?

Prob $(X = 1) = 1/2$

Prob $(X = 2) = 1/2$

Example 2: Outcome of rolling a die – RV: # dots showing



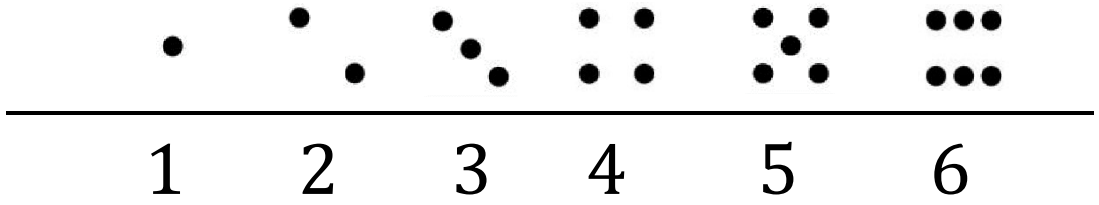
| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|

Possible values of the RV $X$

Prob $(X = x)$ ?

Prob $(X = 1) = 1/6$
Prob $(X = 2) = 1/6$
$\cdots$
Prob $(X = 6) = 1/6$

Example 3: Outcome of rolling a die – RV: # dots odd or even




If # dots odd: $x = 0$
If # dots even: $x = 1$

} Defines the values of the RV $X$

Prob $(X = x)$ ?

Prob $(X = 0) = 3/6 = 1/2$
Prob $(X = 1) = 3/6 = 1/2$

Example 4: Toss a coin three times – RV: total # heads

| | HHH | HHT | HTH | THH | HTT | THT | TTH | TTT |
|---|---|---|---|---|---|---|---|---|
| $x$ | 3 | 2 | 2 | 2 | 1 | 1 | 1 | 0 |

Possible values of the RV $X$

Prob $(X = x)$ ?

Prob $(X = 3) = 1/8$
Prob $(X = 2) = 3/8$
Prob $(X = 1) = 3/8$
Prob $(X = 0) = 1/8$

# Example 5: NAT2 Phenotype* in South African Infants

| | Slow | Inter | Fast |
|---|---|---|---|
| $x$ | 1 | 2 | 3 |

Possible values of the RV $X$

$$\text{Prob}\,(X = 1) = 0.32$$
$$\text{Prob}\,(X = 2) = 0.44$$
$$\text{Prob}\,(X = 3) = 0.24$$

**N*-acetyltransferase type 2 (NAT2) is a liver enzyme that metabolizes (acetylates) many drugs; *NAT2* activity is genetically determined through a single nucleotide polymorphism in the encoding gene, and has a trimodal metabolic phenotype

# Gambler's Fallacy

- Consider the repeated toss of a fair coin $N$ times; RV is # of times heads comes up

- Probability of having the coin come up heads $N$ times in a row = Prob $(X = N) = \frac{1}{2^N}$
  - 1 coin toss: Prob $(X = 1) = \frac{1}{2}$
  - 2 coin tosses: Prob $(X = 2) = \frac{1}{2^2} = \frac{1}{4}$
  - 5 coin tosses: Prob $(X = 5) = \frac{1}{2^5} = \frac{1}{32}$

- Suppose we have just tossed three heads in a row; what is the probability that the next toss will also come up heads?

# Gambler's Fallacy

- Suppose we have just tossed three heads in a row; what is the probability that the next toss will also come up heads?
  - Prob $(X = 1) = \frac{1}{2}$ (the answer is <u>not</u> $\frac{1}{16}$)

- The previous results do not matter: each coin toss is an INDEPENDENT event, so the likelihood of getting a heads from any single toss is 1 out of 2, or $\frac{1}{2}$

- The "Gambler's Fallacy" involves thinking about how unlikely it would be to get four heads in a row (in this case), and losing sight of the INDEPENDENCE of each individual event

# Independent vs. Dependent Events

- 2 independent events: the result of the second event is in no way affected by the result of the first event:
  - Flipping a coin then flipping a coin
  - Flipping a coin then rolling a die
  - Picking a card from a deck, replacing the card and re-shuffling the deck, then picking another card from the deck

- 2 dependent events: the result of the second event IS affected by the result of the first event:
  - Picking a card from a deck, then picking another card without replacing the first (the deck is not the same the second time around)

# Independent vs. Dependent Events

- If $A$ and $B$ are INDEPENDENT events:

  $\text{Prob}(A \text{ and } B) = \text{Prob}(A) \cdot \text{Prob}(B)$

  Example: probability of rolling a six and then rolling an even number on a six-sided die

$$\text{Prob}(A) \cdot \text{Prob}(B) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$$

- If $A$ and $B$ are DEPENDENT events, and $A$ occurs first:

  $\text{Prob}(A \text{ and } B)$

  $= \text{Prob}(A) \cdot \text{Prob}(B, \text{once } A \text{ has occurred})$

  $= \text{Prob}(A) \cdot \text{Prob}(B|A)$

  Example: probability of picking a King and then another King from a deck without replacing the first card

$$\text{Prob}(A) \cdot \text{Prob}(B \mid A) = \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

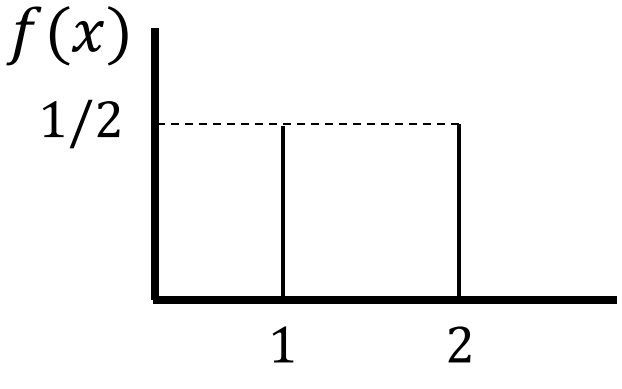
# Probability Function (p.f.)

Let $X$ denote a discrete RV with values: $x_1, x_2, \ldots, x_k$

The Probability Function of $X$ is denoted $f(x)$, where

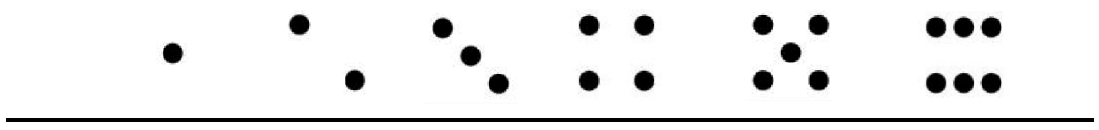
$$f(x) = \text{Prob}(X = x)$$

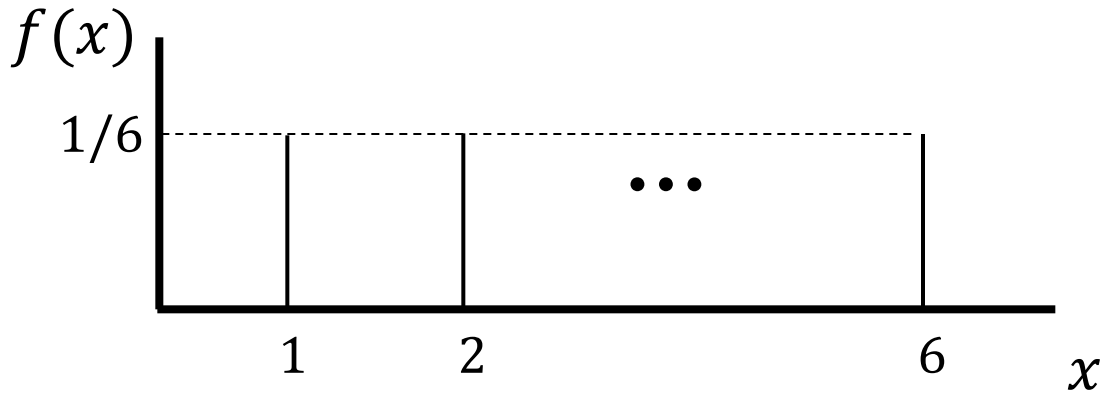
Example 1: Outcome of a coin toss – H $(X = x_1 = 1)$ or T $(X = x_2 = 2)$




$$f(x) = 1/2, \quad x = 1, \; x = 2$$
$$0, \quad \text{elsewhere}$$

Example 2: Outcome of rolling a die – RV: # dots showing

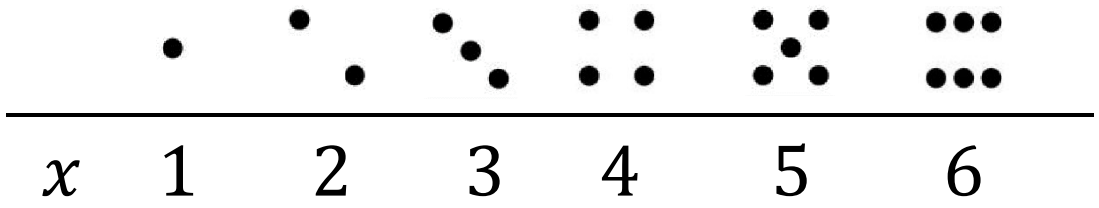


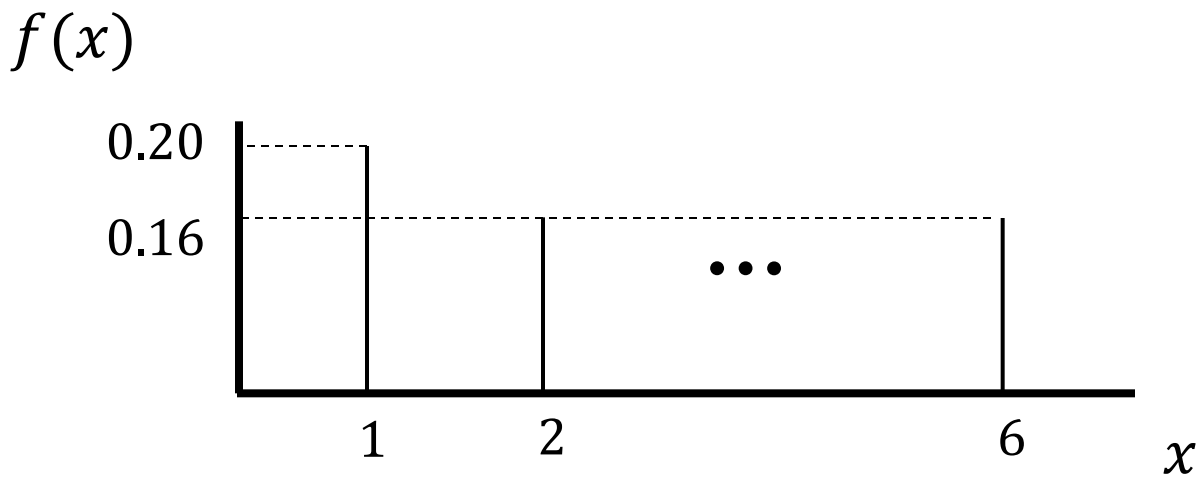




$$f(x) = 1/6, \quad x = 1, 2, \ldots, 6$$
$$0, \quad \text{elsewhere}$$

(Uniform Discrete Probability Function)

Example 3: Outcome of rolling a loaded die – RV: # dots showing




$$f(x) = 0.20, \quad x = 1$$
$$f(x) = 0.16, \quad x = 2, \ldots, 6$$
$$0, \quad \text{elsewhere}$$

Example 4: Toss a coin three times – RV: total # heads

| | HHH | HHT | HTH | THH | HTT | THT | TTH | TTT |
|---|---|---|---|---|---|---|---|---|
| $x$ | 3 | 2 | 2 | 2 | 1 | 1 | 1 | 0 |

Possible values of the RV $X$

Prob $(X = x)$ ?

Prob $(X = 3) = 1/8$
Prob $(X = 2) = 3/8$
Prob $(X = 1) = 3/8$
Prob $(X = 0) = 1/8$

# Example 5: NAT2 Phenotype* in South African Infants

| | Slow | Inter | Fast |
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| $x$ | 1 | 2 | 3 |

Possible values of the RV $X$

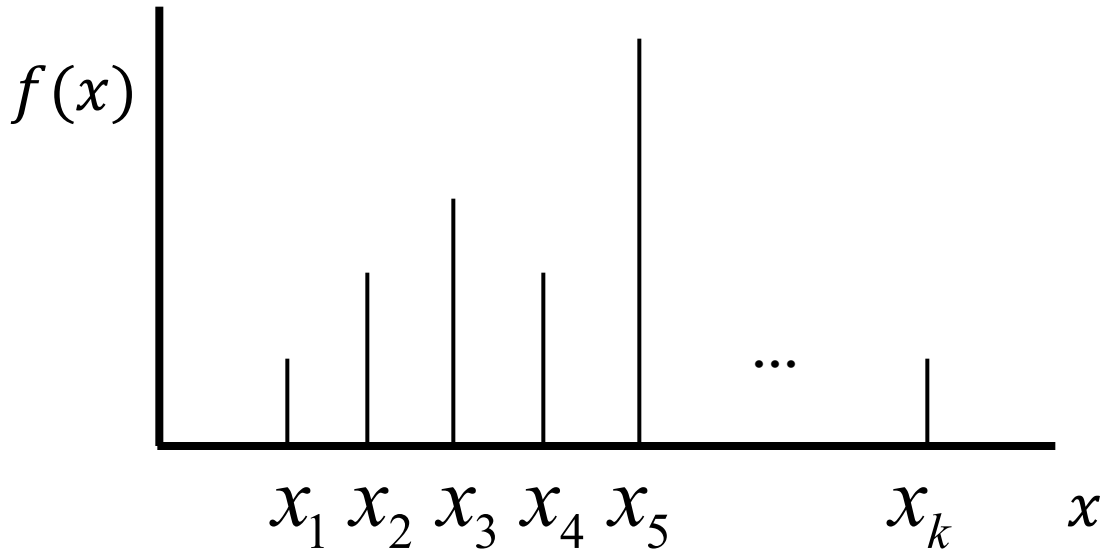
$$\text{Prob}\,(X = 1) = 0.32$$
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**N*-acetyltransferase type 2 (NAT2) is a liver enzyme that metabolizes (acetylates) many drugs; *NAT2* activity is genetically determined through a single nucleotide polymorphism in the encoding gene, and has a trimodal metabolic phenotype

# General Discrete Probability Function (p.f.)

Let $X$ denote a discrete RV with values: $x_1, x_2, \ldots, x_k$

$$f(x) = \text{Prob}(X = x)$$

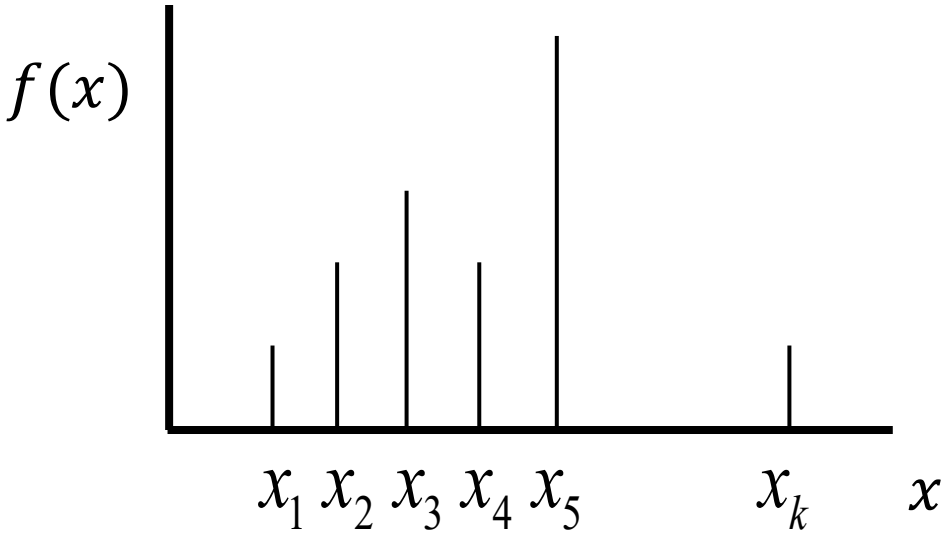



$f(x_i)$ $x_1, x_2, \ldots, x_k$

$f(x) = 0$ elsewhere

# Properties of Discrete Probability Function

$$f(x) = \text{Prob}(X = x)$$




Properties

$$f(x_i) \geq 0$$

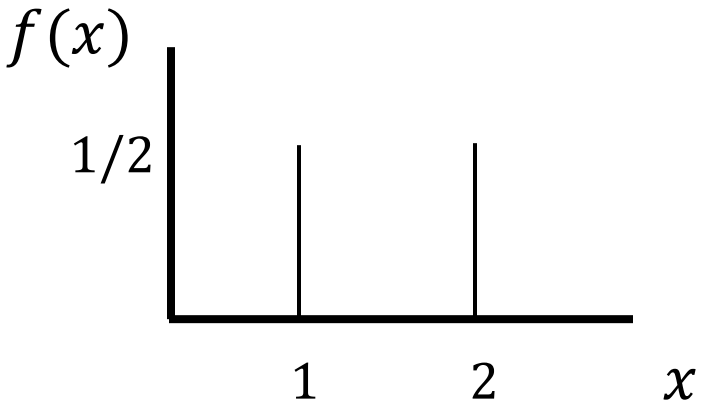
$$\sum_{i=1}^{k} f(x_i) = 1$$

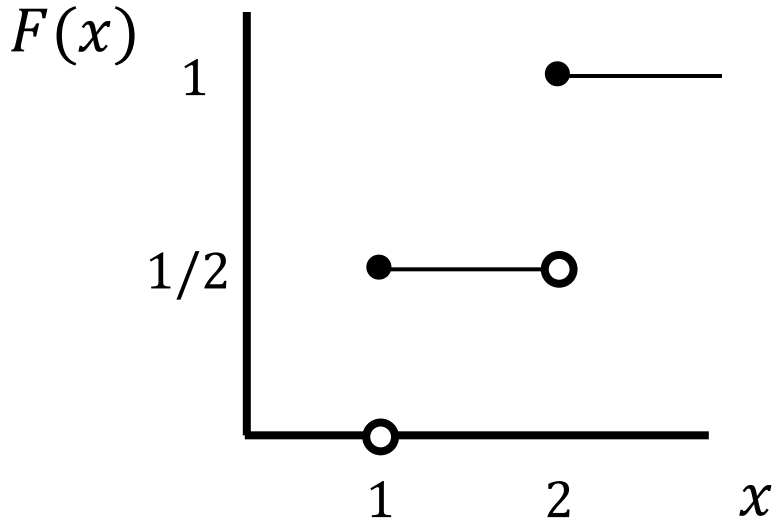
$f(x_i)$ $x_1, x_2, ..., x_k$

$f(x) = 0$ elsewhere

# Distribution Function (d.f.)

$$f(x) = \text{Prob}(X = x) \qquad F(x) = \text{Prob}(X \le x) = \sum_{y:y \le x} f(y)$$




Others: e.g., d.f. for loaded die RV $f(x) = 0.20, \quad x = 1$

$f(x) = 0.16, \quad x = 2, \ldots, 6$

$0, \qquad$ elsewhere

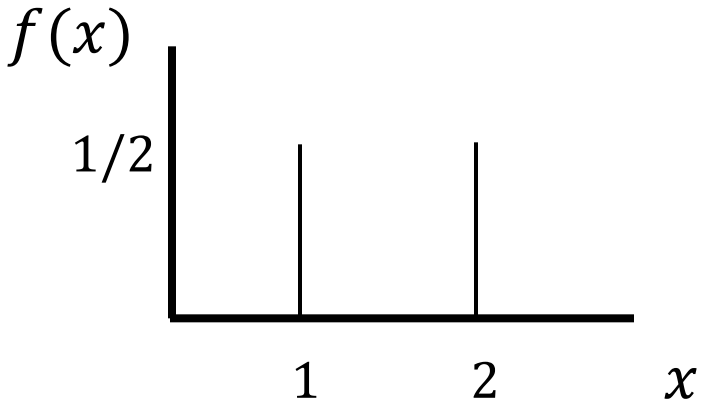
Given the p.f. $f_5(x) = \begin{cases} \frac{x}{6} & \text{for } x = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$ Sketch both the p.f. and d.f.

# Expected Value of $X$ (mean of $X$)

Let $X$ denote a discrete RV with values in set $D$ and p.f. $f(x)$

$$E[X] = \mu = \sum_{x \in D} x \cdot f(x) = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Example




$$E[X] = \sum_{x \in D} x f(x)$$

$$= 1\left(\tfrac{1}{2}\right) + 2\left(\tfrac{1}{2}\right) = 1.5$$



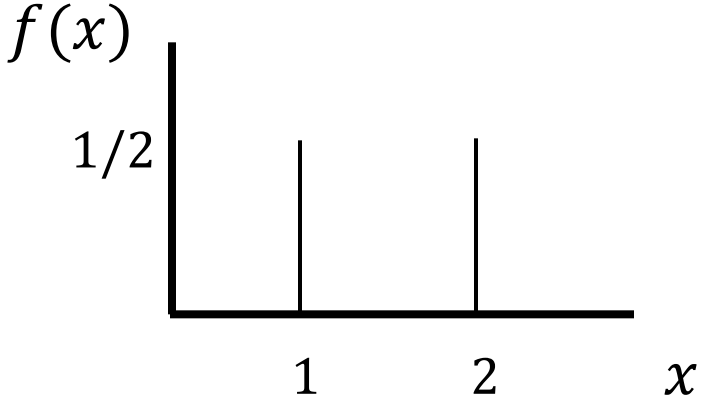
Others: e.g., loaded die

# Expected Value of a Function of $X$

Let $X$ denote a discrete RV with values in set $D$ and p.f. $f(x)$

Define a function of $X$ as $h(X)$

$$E[h(X)] = \sum_{x \in D} h(x) \cdot f(x) = \frac{1}{N} \sum_{i=1}^{N} h(x_i)$$



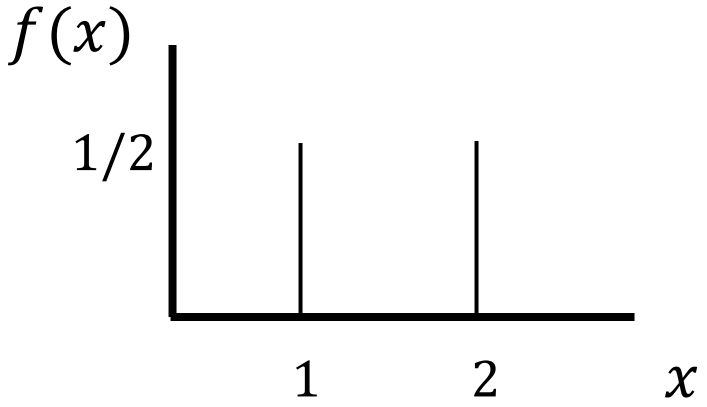

Examples

$$h(X) = X^2$$

$$h(X) = \left(X - E[X]\right)^2 = \left(X - 1.5\right)^2$$

For each, find: $E[h(X)]$

Examples




Find: $E[h(X)]$

$$h(X) = X^2$$

$$h(X) = (X - E[X])^2$$

$$h(X) = X^2: \quad E[X^2] = \sum_{x \in D} x^2 f(x) = 1^2\left(\tfrac{1}{2}\right) + 2^2\left(\tfrac{1}{2}\right) = 2.5$$

$$h(X) = (X - E[X])^2: \quad E\left[(X - E[X])^2\right] = E\left[(X - 1.5)^2\right]$$

**Variance of $X$**

$$= \sum_{x \in D} (x - 1.5)^2 f(x)$$

$$= (-0.5)^2\left(\tfrac{1}{2}\right) + (0.5)^2\left(\tfrac{1}{2}\right) = 0.25$$

Given the p.f. $f_5(x) = \begin{cases} \frac{x}{6} & x=1, 2, 3 \\ 0 & \text{elsewhere} \end{cases}$

Calculate both

$E[X]$

$E\left[\left(X - E[X]\right)^2\right]$

# roulette: expected value example

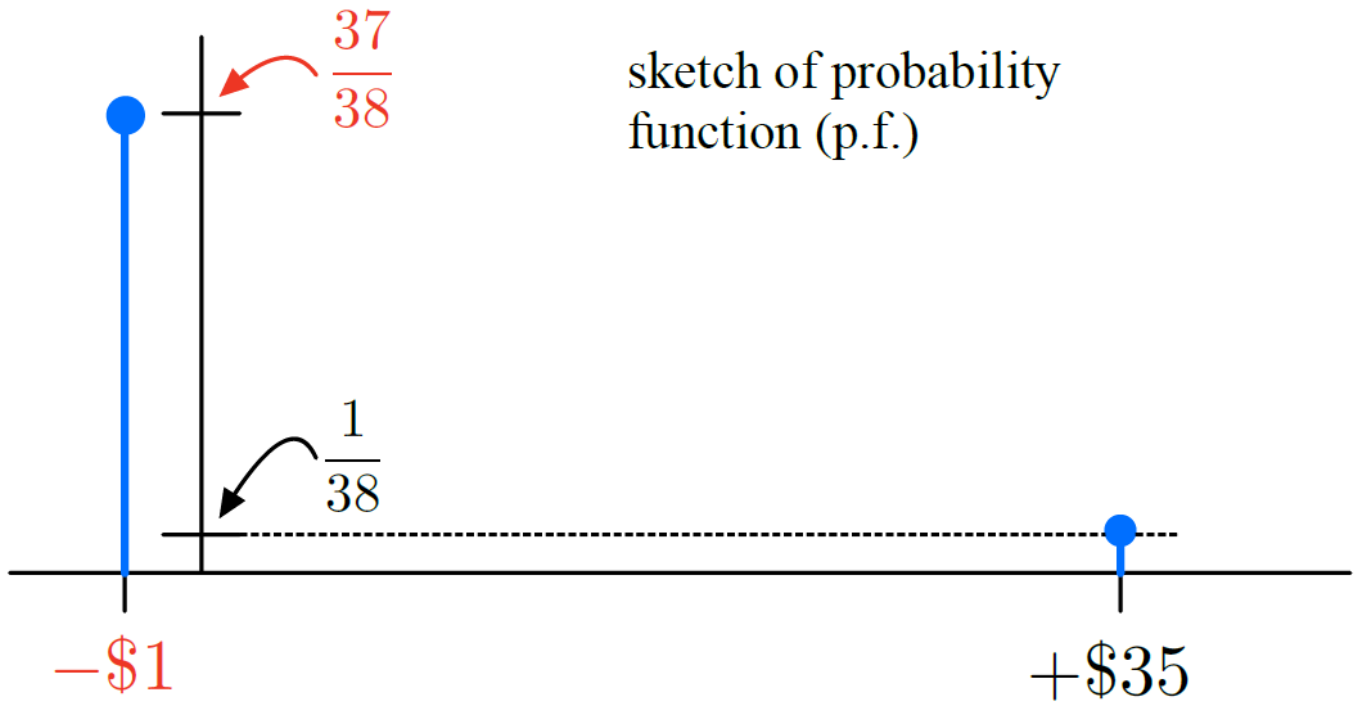
38 numbered pockets, $1 "straight up" bet  Payout is 35:1

## What is the probability function for earnings?

$$f(x) = {}^{37}\!/_{38}, \quad x = -1$$

$$f(x) = {}^{1}\!/_{38}, \quad x = 35$$

$$0, \quad \text{elsewhere}$$




sketch of probability function (p.f.)

# What are the expected earnings from one play?

$$E[X] = \sum_{x \in D} x f(x)$$

$$E[\text{gain from \$1 bet}] = -\$1 \cdot \frac{37}{38} + \$35 \cdot \frac{1}{38}$$

the cost of betting and losing

if you win, you get \$36, so your winnings $= \$36 - \$1 = \$35$

$$= -\$0.0526$$

on average, you are expected to lose approximately 5 cents per \$1 bet